

## Solution Requirements

|                      |                                         |
|----------------------|-----------------------------------------|
| <b>Date</b>          | 18 July 2026                            |
| <b>Team ID</b>       | SWTID-2026-5023                         |
| <b>Project Name</b>  | Rising Waters (Flood Prediction System) |
| <b>Maximum Marks</b> | 4 Marks                                 |

### Define Functional and Non-Functional Requirements:

- **Functional Requirements (FR)** define specific behaviors, functions, or features of the system (What the system should *do*).
- **Non-Functional Requirements (NFR)** specify the criteria that judge the operation of a system, rather than specific behaviors (How the system should *be*).

### Step 1: Functional Requirements (FR)

| S.No | Requirement Category | Requirement Description                                     | Priority (High/Medium/Low) |
|------|----------------------|-------------------------------------------------------------|----------------------------|
| 1    | Authentication       | <i>User can securely access the application.</i>            | High                       |
| 2    | Authorization levels | <i>Only authorized users can use the prediction system.</i> | Medium                     |

|          |                                   |                                                                          |        |
|----------|-----------------------------------|--------------------------------------------------------------------------|--------|
| <b>3</b> | External interfaces               | <i>Accepts flood parameter inputs through the web interface.</i>         | High   |
| <b>4</b> | Transactions processing           | <i>Processes input data and predicts flood risk using the KNN model.</i> | High   |
| <b>5</b> | Reporting                         | <i>Displays flood prediction results on the screen.</i>                  | High   |
| <b>6</b> | Business rules, etc.              | <i>Prediction is based on trained flood dataset values.</i>              | High   |
| <b>7</b> | Compliance to laws or regulations | <i>User data is kept private and used only for prediction.</i>           | Medium |
| <b>8</b> | <i>Other</i>                      | <i>Allows multiple flood predictions without restarting the system.</i>  | Low    |

## Step 2: Non-Functional Requirements (NFR)

| S.No | NFR Category               | Requirement Description                       | Target Metric / Acceptance Criteria |
|------|----------------------------|-----------------------------------------------|-------------------------------------|
| 1    | Performance & Speed        | <i>Generate prediction quickly.</i>           | <i>Response time &lt; 2 seconds</i> |
| 2    | Scalability                | <i>Support multiple users.</i>                | <i>Up to 100 concurrent users</i>   |
| 3    | Security & Data Privacy    | <i>Protect user input data.</i>               | <i>Secure data handling</i>         |
| 4    | Reliability & Availability | <i>System should work consistently.</i>       | <i>99% availability</i>             |
| 5    | Usability & Accessibility  | <i>Easy and user-friendly interface.</i>      | <i>Simple navigation and input</i>  |
| 6    | Other                      | <i>Easy to maintain and update the model.</i> | <i>Easy maintenance</i>             |